Year 11 · AQA Level 2 Further Maths · Covered topic

Algebraic fractions: simplify, add and solve

Revisit and practise · OnMaths · Created by Mr Ali

Today I will learn to...

Goals

1. Simplify algebraic fractions by factorising

2. Add, subtract and solve with algebraic fractions

Do now: retrieval practice

4 mins

1 Simplify 12/18

2/3

2 Work out 1/3 + 1/4

7/12

3 Factorise $x^2 + 5x + 6$

$(x + 2)(x + 3)$

4 Factorise $x^2 - 16$

$(x - 4)(x + 4)$

Objective 1

Simplify algebraic fractions by factorising

Key idea

Learn

!

Factorise the top and the bottom fully, then cancel common FACTORS (whole brackets), never single terms. (x + 3)/(x + 5) does not simplify.

Objective 1: Simplify algebraic fractions by factorising

I do: example 1

I do

Simplify $(x^2 + 5x + 6)/(x^2 - 4)$

1. Top: $(x + 2)(x + 3)$
2. Bottom: $(x - 2)(x + 2)$
3. Cancel the common factor $(x + 2)$
4. $= (x + 3)/(x - 2)$

I do: example 2

I do

Simplify $(2x^2 - 8x)/(x^2 - 16)$

1. Top: $2x(x - 4)$
2. Bottom: $(x - 4)(x + 4)$
3. Cancel $(x - 4)$
4. $= 2x/(x + 4)$

I do: example 3

I do

Simplify $(2x^2 + 5x - 3)/(4x^2 - 1)$

1. Top: $(2x - 1)(x + 3)$
2. Bottom: $(2x - 1)(2x + 1)$
3. Cancel $(2x - 1)$
4. $= (x + 3)/(2x + 1)$

Mini whiteboards

We do

1. Simplify $(x^2 - 9)/(x^2 + x - 6)$

Answer: $(x - 3)/(x - 2)$

2. Simplify $(3x + 12)/(x^2 + 4x)$

Answer: $3/x$

3. Simplify $(x^2 - 1)/(x^2 + 3x + 2)$

Answer: $(x - 1)/(x + 2)$

4. Simplify $(4x - 8)/(x^2 - 4)$

Answer: $4/(x + 2)$

You do: choose your challenge

You do

Bronze

Simplify $(x^2 + 3x)/(x^2 - 9)$

Silver

Simplify $(x^2 - 2x - 15)/(x^2 - 25)$

Gold

Simplify $(2x^2 - x - 1)/(x^2 - 1)$

Platinum

Simplify $(6x^2 + x - 2)/(9x^2 - 4)$

You do: answers

Check

Bronze

$\frac{x}{x-3}$

Silver

$\frac{x+3}{x+5}$

Gold

$\frac{2x+1}{x+1}$

Platinum

$\frac{2x-1}{3x-2}$

If you finish early

Extra

Bronze

Simplify $(x^2 + 2x)/(x^2 - 4)$

$x/(x - 2)$

Silver

Simplify $(x^2 + 6x + 8)/(x^2 + 3x - 4)$

$(x + 2)/(x - 1)$

Gold

Simplify $(3x^2 - 5x - 2)/(x^2 - 4)$

$(3x + 1)/(x + 2)$

Platinum

Simplify $(4x^2 - 25)/(6x^2 + 13x - 5)$

$(2x - 5)/(3x - 1)$

Exam question 1

3 marks

Simplify fully $(x^2 - x - 12)/(x^2 - 16)$

Space for working

Mark scheme: Top $(x - 4)(x + 3)$ (M1). Bottom $(x - 4)(x + 4)$ (M1). $(x + 3)/(x + 4)$ (A1)

Exam question 2

3 marks

Simplify fully $(3x^2 - 12)/(x^2 + 3x - 10)$

Space for working

Mark scheme: Top $3(x - 2)(x + 2)$ (M1). Bottom $(x + 5)(x - 2)$ (M1). $3(x + 2)/(x + 5)$ (A1)

Objective 1 achieved

Simplify algebraic fractions by factorising



Objective 2

Add, subtract and solve with algebraic fractions

✓ Achieved: Simplify algebraic fractions by factorising

Key idea

Learn

!

To add or subtract, use a common denominator: multiply each fraction top and bottom by the other denominator. To solve an equation, multiply every term by the common denominator to clear the fractions.

Objective 2: Add, subtract and solve with algebraic fractions

I do: example 1

I do

Write $2/x + 3/(x + 1)$ as a single fraction

1. Common denominator: $x(x + 1)$
2. $2(x + 1) + 3x$ on the top
3. $2x + 2 + 3x = 5x + 2$
4. $= (5x + 2)/(x(x + 1))$

I do: example 2

I do

Write $4/(x - 2) - 1/(x + 3)$ as a single fraction

1. Common denominator: $(x - 2)(x + 3)$
2. $4(x + 3) - (x - 2)$ on the top
3. $4x + 12 - x + 2 = 3x + 14$
4. $= (3x + 14)/((x - 2)(x + 3))$

I do: example 3

I do

Solve $6/x - 4/(x + 1) = 1$

1. Multiply every term by $x(x + 1)$
2. $6(x + 1) - 4x = x(x + 1)$
3. $2x + 6 = x^2 + x$, so $x^2 - x - 6 = 0$
4. $(x - 3)(x + 2) = 0$: $x = 3$ or $x = -2$

Mini whiteboards

We do

1. Write $1/x + 1/(2x)$ as a single fraction

Answer: $3/(2x)$

2. Write $2/(x + 1) + 1/(x - 1)$ as a single fraction

Answer: $(3x - 1)/((x + 1)(x - 1))$

3. Write $3/x - 1/(x + 2)$ as a single fraction

Answer: $(2x + 6)/(x(x + 2))$

4. Solve $2/x + 1 = 3/x$

Answer: $x = 1$

You do: choose your challenge

You do

Bronze

Write $3/x + 2/x^2$ as a single fraction

Silver

Write $5/(x + 2) - 2/(x + 1)$ as a single fraction

Gold

Solve $4/(x + 1) = 3/(x - 2)$

Platinum

Solve $1/x + 2/(x + 3) = 1/2$

You do: answers

Check

Bronze

$(3x + 2)/x^2$

Silver

$(3x + 1)/((x + 2)(x + 1))$

Gold

$4x - 8 = 3x + 3$, so $x = 11$

Platinum

$2(x + 3) + 4x = x(x + 3)$, $x^2 - 3x - 6 = 0$, $x = 4.37$ or $x = -1.37$

If you finish early

Extra

Bronze

Write $1/x + 1/(x + 1)$ as a single fraction

$(2x + 1)/(x(x + 1))$

Silver

Write $3/(x - 1) - 2/(x + 2)$ as a single fraction

$(x + 8)/((x - 1)(x + 2))$

Gold

Solve $5/(x + 2) = 2/(x - 1)$

$5x - 5 = 2x + 4$, $x = 3$

Platinum

Solve $4/x + 3/(x - 1) = 3$. Give answers to 2 dp.

$3x^2 - 10x + 4 = 0$, $x = 2.87$ or $x = 0.46$

Exam question 1

3 marks

Write $\frac{3}{(x + 2)} + \frac{5}{(x - 4)}$ as a single fraction in its simplest form

Space for working

Mark scheme: $3(x - 4) + 5(x + 2)$ (M1) over $(x + 2)(x - 4)$ (M1). $\frac{(8x - 2)}{((x + 2)(x - 4))}$ (A1)

Exam question 2

4 marks

Solve $2/(x - 1) + 3/(x + 1) = 1$

Space for working

Mark scheme: $2(x + 1) + 3(x - 1) = (x - 1)(x + 1)$ (M1). $5x - 1 = x^2 - 1$ (M1). $x^2 - 5x = 0$ (M1). $x = 0$ or $x = 5$ (A1)

Objective 2 achieved

Add, subtract and solve with algebraic fractions



Real life

Apply

Sam cycles 12 km at x km/h, then another 12 km at (x + 2) km/h. Write the total time as a single fraction.

$12/x + 12/(x + 2) = (24x + 24)/(x(x + 2))$ hours

Extension: stretch yourself

Stretch

1. Simplify $(x^2 - 1)/(x^2 + 2x + 1) \times (x + 1)/(x - 1)$

$(x - 1)(x + 1)/(x + 1)^2 \times (x + 1)/(x - 1) = 1$

2. Solve $x/(x + 2) - 1/(x - 1) = 1$

$x(x - 1) - (x + 2) = (x + 2)(x - 1)$, so $x^2 - 2x - 2 = x^2 + x - 2$ and $x = 0$

Exit ticket: exam questions

Exit

1. Simplify fully $(x^2 - 4)/(x^2 + 5x + 6)$ **(2 marks)**

2. Write $2/x + 3/(x + 4)$ as a single fraction **(2 marks)**

3. Solve $3/x + 4/(x + 1) = 2$ **(3 marks)**

Exit ticket: mark scheme

Mark it

Q1 $(x - 2)(x + 2)/((x + 2)(x + 3))$ (M1) $= (x - 2)/(x + 3)$ (A1) [2]

Q2 $(2(x + 4) + 3x)/(x(x + 4))$ (M1) $= (5x + 8)/(x(x + 4))$ (A1) [2]

Q3 $3(x + 1) + 4x = 2x(x + 1)$ (M1). $2x^2 - 5x - 3 = 0$ (M1). $x = 3$ or $x = -1/2$ (A1) [3]

Total: ____ / 7

Today I achieved

- ✔ **Simplify algebraic fractions by factorising**
- ✔ **Add, subtract and solve with algebraic fractions**

Rate yourself: 1 = still unsure 2 = getting there 3 = confident

Homework: exam questions

Due next lesson

1. Simplify fully $(x^2 + 7x + 10)/(x^2 - 25)$ [2]

2. Simplify fully $(4x^2 - 9)/(2x^2 + x - 3)$ [3]

3. Write $1/(x - 3) + 2/(x + 5)$ as a single fraction [2]

4. Write $x/(x + 1) - 2/(x - 2)$ as a single fraction [3]

5. Solve $5/(x + 2) + 2/x = 1$. Give your answers to 2 dp. [3]

Extra practice: Mr Barton Maths: AQA Further Maths → search "Algebraic fractions"